DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING (AI & ML, IOT), RAI

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD

B.TECH. II YEAR

CSE (INTERNET OF THINGS)

III SEMESTER **R25**

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT204	Probability and Statistical Analysis	2	1	0	3	3
25PC1CY201	Computer Networks	3	0	0	3	3
25PC1CS201	Object Oriented Programming Through JAVA	3	0	0	3	3
25PC1IT201	Database Management Systems	3	0	0	3	3
25HS1MG201	Innovation and Entrepreneurship	2	0	0	2	2
25BS2MT211	Computational Mathematics Laboratory	0	0	2	2	1
25PC2CY201	Computer Networks Laboratory	0	0	2	2	1
25PC2CS201	Object Oriented Programming Through JAVA Laboratory	0	0	2	2	1
25PC2IT201	Database Management Systems Laboratory	0	0	2	2	1
25PC2IN211	IoT Readiness Laboratory	0	0	2	2	1
25SD5CS201	Data Visualization Laboratory	0	0	2	2	1
25MN6HS103	Happiness and Wellbeing	2	0	0	2	0
Total		**15**	**1**	**12**	**28**	**20**

IV SEMESTER **R25**

Course Code	Title of the Course	L	T	P/D	CH	C
25PC1AM201	Machine Learning	3	0	0	3	3
25PC1CS205	Design and Analysis of Algorithms	3	0	0	3	3
25PC1AM101	Mathematical Foundations for Computer Science	3	0	0	3	3
25PC1IT204	Operating Systems	3	0	0	3	3
25PC1IT208	Computer Organization and Architecture	3	0	0	3	3
25PW4IN201	Design Thinking	1	0	2	3	2
25PC2AM201	Machine Learning laboratory	0	0	2	2	1
25PC2AM211	Operating Systems with Linux Laboratory	0	0	2	2	1
25SD5CS202	Web Application Development	0	0	2	2	1
25VA7HS201	Ancient Wisdom	1	0	0	1	1
Total		**17**	**0**	**8**	**25**	**21**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25BS1MT204) PROBABILITY AND STATISTICAL ANALYSIS

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: First Year Mathematics courses

COURSE OBJECTIVES:

- To learn the theory of Random Variable, and probability distributions of single random variables
- To understand the concept of sampling distribution
- To know the various methods to test the hypothesis for large and small samples
- To find the linear and polynomial relationship between the given data set
- To understand the concepts of Multivariable linear Regression model

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the concepts of Random variable and distributions to some case studies
CO-2: Evaluate statistical parameters of continuous probability distributions and sampling distributions
CO-3: Apply the knowledge to test the hypotheses for large and small samples
CO-4: Use and fit a linear and polynomial regression model to the data and use it for prediction
CO-5: Model the linear relationship between the explanatory (independent) Variables and response(dependent)variables

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	2	1	-	-	-	-	-	-	3	1	2
CO-2	3	2	1	2	1	-	-	-	-	-	-	3	1	2
CO-3	3	2	1	2	1	-	-	-	-	-	-	3	2	2
CO-4	3	3	3	2	2	-	-	-	-	-	-	3	2	2
CO-5	3	3	2	2	2	-	-	-	-	-	-	3	2	2

UNIT-I:
Random Variables and Probability Distributions: Concept of a Random Variable – Discrete Probability Distributions – Continuous Probability Distributions – Mean of a Random Variable – Variance of a Random Variable
Discrete Probability Distributions: Binomial Distribution – Poisson distribution

UNIT-II:
Continuous Distributions and sampling: Uniform Continuous Distribution – Normal Distribution – Areas under the Normal Curve – Applications of the Normal Distribution

Sampling Distributions: Population, sample, parameter, statistic, Types of Sampling – Sampling Distributions with and without replacement – Sampling Distribution of Means – Central Limit Theorem.
Introduction – Statistical Inference – Classical Methods of Estimation – Single Sample: Estimating the mean – Standard error of a point estimate - Estimating a proportion.

UNIT-III:
Tests of Hypothesis:
Statistical Hypothesis: General Concepts (Types of hypothesis, Type-I & II errors, Critical region, One-tailed and two-tailed) – Testing a Statistical Hypothesis-small (t-test) and large samples. Single sample: Tests concerning a single mean. Two samples: Tests on two mean. One sample: Test on a single proportion. Two samples: Tests on two proportions. F-test for equality of variances..

UNIT-IV:
Applied Statistics: Correlation and Regression: Curve fitting by the method of least squares, fitting of straight lines, second degree parabolas and more general curves, Correlation coefficient, Rank correlation

UNIT-V:
Multivariate Analysis: Multivariate Analysis: Multivariate Normal Distribution Function, Properties of Multivariate Normal Distribution, Conditional Distribution and its relation to regression model, Linear Regression Analysis-Step up, Step down (Forward/ Backward), Subset and Best Model (Step-wise), Tenfold Validation, One -Off technique, (Leave-Out), Multi Collinearity, Variation Inflation Factors, Principal Component Analysis (PCA) based regression analysis.

TEXT BOOKS:

1. Probability & Statistics for Engineers & Scientists, Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, Keying Ye, 9th Edition, Pearson Publishers
2. Fundamentals of Mathematical Statistics, S. C. Gupta and V. K. Kapoor, Khanna publications
3. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 36th Edition, 2010

REFERENCES:

1. Fundamentals of Probability and Statistics for Engineers, T. T. Soong, John Wiley & Sons Ltd., 2004
2. Probability and Statistics for Engineers and Scientists, Sheldon M. Ross, Academic Press

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC1CY201) COMPUTER NETWORKS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Elements of CSE

COURSE OBJECTIVES

- To develop an understanding of modern network architectures from a design and performance perspective
- To introduce the major concepts and principles involved in the data link layer and network layer
- To learn how to maintain Quality of Service (QoS) and congestion control in networks
- To understand application layer functionalities and the importance of network security
- To analyze application layer protocols, network services, and basic security mechanisms used in modern computer networks

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand modern network architectures from a design and performance perspective

CO-2: Learn major concepts and principles involved in the data link layer and network layer

CO-3: Analyze methods to maintain QoS and congestion control in networks

CO-4: Differentiate between TCP and UDP protocols and identify techniques to improve QoS

CO-5: Understand application layer functionalities and the importance of network security

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	2	-	-	-	-	-	1	3	2	1
CO-2	3	3	2	1	2	-	-	-	-	-	1	3	2	2
CO-3	2	3	3	2	2	1	-	-	-	-	2	3	2	3
CO-4	2	3	2	1	3	-	-	-	1	-	2	3	2	2
CO-5	2	2	2	1	2	2	2	-	2	-	3	3	3	3

UNIT-I:

Data Communication Components, Representation of data and its flow, Computer Networks, Network Topologies, Protocols and Standards, OSI Model, Transmission Media, LAN Overview (Wired LAN, Wireless LANs), Connecting LANs and Virtual LANs, Bandwidth Utilization Techniques – Multiplexing (FDM, TDM, WDM), Spread Spectrum Concepts.

UNIT-II:
Data Link Layer and Medium Access Sub Layer, Error Detection and Error Correction Fundamentals, Block Coding, Hamming Distance, CRC, Flow Control and Error Control Protocols – Stop and Wait, Go-Back-N ARQ, Selective Repeat ARQ, Sliding Window, Piggybacking, Random Access, Multiple Access Protocols – Pure ALOHA, Slotted ALOHA, CSMA/CD, CDMA/CA.

UNIT-III:
Network Layer Concepts, Switching Techniques, Logical Addressing – IPv4 and IPv6, Address Mapping – ARP, RARP, BOOTP, DHCP, Delivery, Forwarding, and Unicast Routing Protocols.

UNIT-IV:
Transport Layer Concepts, Process-to-Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), Stream Control Transmission Protocol (SCTP), Congestion Control, Quality of Service (QoS), QoS Improving Techniques – Leaky Bucket and Token Bucket Algorithms.

UNIT-V:
Application Layer Concepts and Protocols, Domain Name System (DNS), Dynamic DNS (DDNS), TELNET, Electronic Mail, File Transfer Protocol (FTP), World Wide Web (WWW), Hypertext Transfer Protocol (HTTP), Simple Network Management Protocol (SNMP), Firewalls.

TEXT BOOKS:

1. Data Communication and Networking, Behrouz A. Forouzan, 4th Edition, Tata McGraw-Hill
2. Computer Networks, Andrew S. Tanenbaum, 8th Edition, Pearson

REFERENCES:

1. Data and Computer Communication, William Stallings, 8th Edition, Pearson
2. Internetworking with TCP/IP, Volume 1, Douglas Comer, 6th Edition, PHI
3. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley

ONLINE RESOURCES:
1. NPTEL Course Link:
https://www.youtube.com/playlist?list=PLbMVogVj5nJQ3EZa7mBWKaKOJqvS9UE0G

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC1CS201) OBJECT ORIENTED PROGRAMMING THROUGH JAVA

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand core principles of Object-Oriented Programming and Java basics to solve programming problems effectively
- To apply event-driven programming, multithreading, and exception handling to develop robust Java applications
- To analyze program structures and data using collections, generics, and wrapper classes for efficient data management
- To design GUI applications with Swing components and basic JDBC connectivity for real-world scenarios

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Demonstrate the use of classes, objects, recursion, and nested classes to build modular programs

CO-2: Implement inheritance, interfaces, and polymorphism to create reusable and extensible Java applications

CO-3: Develop programs handling exceptions and utilizing multithreading to achieve concurrent execution

CO-4: Construct interactive GUI applications using Swing, layout managers, and event handling mechanisms

CO-5: Employ collections, generics, and JDBC to efficiently store, retrieve, and manipulate data in Java programs

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	2	-	-	-	-	-	1	3	2	2
CO-2	3	3	3	-	2	-	-	-	-	-	1	3	3	2
CO-3	3	3	2	2	2	-	-	-	-	-	2	2	3	3
CO-4	2	2	3	1	3	-	-	1	2	-	1	2	3	3
CO-5	3	3	2	1	3	-	-	-	-	-	2	3	3	3

UNIT-I:

Object-Oriented Thinking: Need for OOP paradigm, summary of OOP concepts, coping with complexity, abstraction mechanisms, Data types, variables, scope and lifetime of variables, arrays, operators, expressions, control statements, Type conversion and casting,

Java Basics: simple Java program, Concepts of classes, objects, constructors, methods, access control, this keyword, garbage collection, Method and constructor overloading, recursion, Nested and inner classes, Exploring String class, Type Wrappers

UNIT-II:

Inheritance: Hierarchical abstractions; forms of inheritance: specialization, specification, construction, extension, limitation, combination, Member access rules, super keyword uses, using final keyword with inheritance

Polymorphism: method overriding, abstract classes, the Object class

Packages: Defining, creating, and accessing packages; understanding CLASSPATH; importing packages

Interfaces: Differences between classes and interfaces; defining an interface; implementing and applying interfaces; variables in interface; extending interfaces

Generics: Introduction to Generics – generic classes, generic methods

UNIT-III:

Exception Handling: Concepts and benefits of exception handling, Exception hierarchy; usage of try, catch, throw, throws, finally, Built-in exceptions; creating own exception subclasses.

Multithreading: Differences between multithreading and multitasking, Thread life cycle, creating threads, thread priorities, Synchronizing threads, inter-thread communication, Thread groups, daemon threads.

UNIT-IV:

Exploring String class and Object class, Exploringjava. util package: Introduction to Collections Framework: List, Set, Map – basic usage, Exploring java. io package

Event Handling: Events, event sources, event classes, event listeners, delegation event model, Handling mouse and keyboard events, adapter classes, Layout manager types – border, grid, flow, card, grid bag

UNIT-V:

Swing: Introduction to Swing, limitations of AWT, MVC architecture,

Components and containers: JFrame, JComponent, JLabel, ImageIcon, JTextField, JButton, JCheckBox, JRadioButton, JList, JComboBox, Trees, Tables, Menu basics, Menu-related classes – JMenuBar, JMenu, JMenuItem, JCheckBoxMenuItem, JRadioButtonMenuItem, JSeparator, Creating a popup menu

JDBC: Types of Drivers, JDBC Architecture, Basic steps in developing JDBC applications, JDBC classes and Interfaces, Creating a new database and table with JDBC.

TEXT BOOKS:

1. Java The Complete Reference, 13th Edition, Herbert Schildt, Dr. Denny Coward, Mc Graw Hill
2. Understanding OOP with Java, Updated Edition, T. Budd, Pearson Education

REFERENCES:

1. An Introduction to Programming and OO Design using Java, J. Nino and F. A. Hosch, John Wiley & Sons
2. An Introduction to OOP, 3rd Edition, T. Budd, Pearson Education
3. Introduction to Java Programming, Y. Daniel Liang, Pearson Education
4. An Introduction to Java programming and Object-Oriented Application Development, R. A. Johnson, Thomson
5. Core Java 2, Vol 1, Fundamentals, Cay. S. Horstmann and Gary Cornell, 8th Edition, Pearson Education

ONLINE RESOURCES:

1. https://docs.oracle.com/javase/tutorial/java/concepts/

2. https://www.javatpoint.com.cach3.com/java-what-where-and-why.html

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC1IT201) DATABASE MANAGEMENT SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Data Structures

COURSE OBJECTIVES:

- To understand basic concepts and applications of database management systems
- To understand concepts of ER model, relational model and model the database for the given scenarios and schema refinement through normalization
- To master the concepts of SQL, relational algebra, relational calculus and construct queries
- To introduce transaction management concepts, concurrency control protocols, recovery mechanism and the concepts of data storage

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Gain knowledge on underlying concepts of database system architecture and design ER models for real-world application scenarios

CO-2: Convert the ER-model to relational tables and perform schema refinement using Normalization

CO-3: Formulate queries in SQL, Relational Algebra and Relational Calculus

CO-4: Relate and apply the concept of transaction management, concurrency control and recovery in database

CO-5: Familiarize with basic database storage structures and access techniques

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	-	1	-	-	-	-	-	1	2	3	1
CO-2	3	3	2	1	1	-	-	-	-	-	1	2	3	1
CO-3	3	3	2	1	2	-	-	-	-	-	1	2	3	1
CO-4	3	3	2	2	2	1	-	-	-	-	2	2	3	1
CO-5	3	2	1	1	2	-	-	-	-	-	2	2	3	1

UNIT-I:

Introduction to DBMS: A Historical Perspective, File Systems versus a DBMS, Data Models, Levels of Abstraction in a DBMS, Data Independence, Structure of a DBMS.

Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes, and Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design With the ER Model.

UNIT-II:

Introduction to the Relational Model: Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design. Relational Algebra, Tuple relational Calculus, Domain relational calculus.

UNIT-III:

SQL: DDL commands, DML Commands, Basic Structure of SQL query, NULL values, Set Operations, Aggregate Operations, Join Operations, Sub Queries and Correlated Queries, Introduction to Views, Destroying/Altering Views, Window Functions: RANK, **DENSE_RANK, LEAD, LAG, Common Table Expressions (CTEs), SQL Security:** SQL injection.

PL/SQL: PL/SQL Introduction, Basic Structure, Functions, Triggers, Cursors and Procedures.

Functional Dependencies: Introduction, Basic Definitions, Trivial and Non Trivial Dependencies, Closure of a Set of Dependencies, Closure of Attributes, Irreducible Set of Dependencies.

Schema Refinement: Problems caused by redundancy, decompositions, problems related to decomposition, lossless join decomposition, Dependency Preserving Decomposition, FIRST, SECOND, THIRD Normal Forms, BCNF, Multivalued Dependencies, FOURTH Normal Form, FIFTH Normal Form.

UNIT-IV:

Transaction Management: Transaction Concept, Transaction States, ACID properties, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Testing for Serializability, Recoverability, Implementation of Isolation.

Concurrency Control: Lock Based Protocols, Timestamp Based Protocols, Validation-Based Protocols, Multiple Granularity, Dead Lock Handling.

Recoverability: Failure Classification, Storage Structure, Recovery and Atomicity, Log–Based Recovery, Recovery with Concurrent Transactions, Checkpoints.

UNIT-V:

File Organization: Organization of records in file, Comparison of File Organizations, Data Dictionary Storage.

Indexing and Hashing: Basic Concepts, Primary and Secondary Indices, Cluster Indices, B tree Index files, B+ tree index files, Static Hashing, Dynamic Hashing – Comparison of Indexing with Hashing.

TEXT BOOKS:

1. Database System Concepts, Silberschatz, Korth, 5th Edition, McGraw-Hill, 2005
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata McGraw-Hill, 2002

REFERENCES:

1. Database Systems Design, Implementation, and Management, Peter Rob & Carlos Coronel, 7th Edition, 2006
2. Fundamentals of Database Systems, Elmasri Navathe, Pearson Education, 2017
3. Introduction to Database Systems, C. J. Date, Pearson Education, 1999

4. Database Systems Using Oracle: A Simplified Guide to SQL and PL/SQL, Shah, PHI, 2016
5. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition, 2011

ONLINE RESOURCES:

1. NOC: Data Base Management System, IIT Kharagpur, NPTEL: https://nptel.ac.in/courses/106105175
2. Scalar: https://www.scaler.com/topics/course/dbms/
3. https://www.w3schools.com/sql/default.asp

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25HS1MG201) INNOVATION AND ENTREPRENEURSHIP

TEACHING SCHEME		
L	T/P	C
2	0	2

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To familiarize on the basic concepts of innovation, entrepreneurship and its importance
- To identify and analyze the process of problem-opportunity identification, market segmentation, and idea generation techniques
- To initiate prototype development and understand minimum viable product
- To develop initial Business and financial planning and Go-to-Market strategies
- To impart knowledge on establishing startups, venture pitching and IPR

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the entrepreneurship and the entrepreneurial process and its significance in economic development

CO-2: Assess the problem from an industry perspective and generate solutions using the design thinking principles

CO-3: Assess market competition, estimate market size, and develop a prototype

CO-4: Analyze Business and financial planning models and Go-to-Market strategies

CO-5: Able to build a start-up, register IP and identify funding opportunities

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	-	1	-	-	1	1		-	-	1
CO-2	2	3	2	1	-	1	-	-	2	1		-	1	-
CO-3	1	2	3	2	-	1	-	-	3	2	1	-	1	-
CO-4	1	1	3	3	-	1	-	-	2	3	1	-	1	-
CO-5	1	1	-	-	2	2	-	-	2	2	1	-	-	-

UNIT-I:

Fundamentals of Innovation and Entrepreneurship

Innovation: Introduction, need for innovation, Features, Types of innovations, innovations in manufacturing and service sectors, fostering a culture of innovation, planning for innovation.

Entrepreneurship: Introduction, types of entrepreneurship attributes, mindset of entrepreneurial and intrapreneurial leadership, Role of entrepreneurs in economic development. Women Entrepreneurship, Importance of on-Campus Startups. Understanding to build an entrepreneurial mindset, attributes, and networks in individuals while on campus.

Core Teaching Tool: Simulation, Game, Industry Case Studies (Personalized for students – 16 industries to choose from), Venture Activity.

UNIT-II:

Problem and Customer Identification: Identification of gap, problem, analyzing the problem from an industry perspective, real-world problems, market and customer segmentation, validation of customer problem fit, Iterating problem-customer fit, Competition and Industry trends mapping, and assessing initial opportunity, Porter's Five Force Model. Idea generation, Ideation techniques: Brainstorming, Brain writing, Round robin, and SCAMPER, Design thinking principles, Mapping of solution to problem.

Core Teaching Tool: Several types of activities, including: Class, game, Gen AI, 'Get out of the Building', and Venture Activity.

UNIT-III:

Opportunity Assessment and Prototype Development: Identify and map global competitors, review industry trends, and understand market sizing: TAM, SAM, and SOM. Assessing scope and potential scale for the opportunity.

Understanding prototyping and Minimum Viable Product (MVP). Developing a prototype: Testing and validation.

Core Teaching Tool: Venture Activity, no-code Innovation tools, Class activity

UNIT-IV:

Business & Financial Models: Introduction to Business Model and types, Lean Canvas Approach: 9-block lean canvas model, building a lean canvas for your startup. Business planning: components of Business plan- Sales plan, People plan, and financial plan. Financial Planning: Types of costs, preparing a financial plan for profitability using a financial template, understanding the basics of Unit economics, Economies of Scale and analyzing financial performance. Go-To-Market (GTM) approach – Selecting the Right Channel, creating digital presence, and building a customer acquisition strategy.

Core Teaching Tool: Founder Case Studies – Sama and Securely Share; Class activity and discussions; Venture Activities.

UNIT-V:

Startups and IPR: Startup requirements, building founding team members and mentors, pitch preparation, start-up registration process, funding opportunities and schemes, institutional support to entrepreneurs, startup lifecycle, documentation, legal aspects in startup, venture pitching readiness, National Innovation Startup Policy (NISP) and its features.

Patents, Designs, Patentability, Procedure for grants of patents.indian Scenario of Patenting, International Scenario: International cooperation on Intellectual Property. Patent Rights: Scope of Patent Rights. Copyright, trademark, and GI. Licensing and transfer of technology.

Core Teaching Tool: Expert talks; Cases; Class activity and discussions; Venture Activities.

TEXT BOOKS:

1. Innovation and Entrepreneurship, John R Bessant, Joe Tidd, 4th Edition, Wiley, Latest Edition
2. The Stratup Launch Book - A Practical Guide for Launching Customer Centric Ventures, Ajay Batra, Wiley, 2020
3. Entrepreneurship Development and Small Business Enterprises, Poornima M Charantimath, 3rd Edition, Pearson, 2018

REFERENCES:

1. Entrepreneurship Simplified: From Idea to IPO, Ashok Soota and S. R. Gopalan, Penguin Portfolio, 2021
2. How to Be a Founder: How Entrepreneurs Can Identify, Fund and Launch Their Best Ideas, Alice Bentinck and Matt Clifford, Bloomsbury Business, 2024
3. Scaling Up: How a Few Companies Make It... and Why the Rest Don't, Verne Harnish, Sanage Publishing, 2024
4. Funding Your Startup and Other Nightmares, Dhruv Nath and Sushanto Mitra, Penguin Portfolio, 2020
5. NISP -Brochure inside pages - startup_policy_2019.pdf

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25BS2MT211) COMPUTATIONAL MATHEMATICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Matrices, Iterative Methods and Ordinary Differential Equations

COURSE OBJECTIVES:
- To solve problems of Eigen values and Eigen Vectors using MATLAB
- To solution of algebraic and transcendental equations using MATLAB
- To solve problems of linear system of equations
- To solve the first-order ODEs and higher order linear differential equations with constant coefficients

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Develop the code to find the Eigen values and Eigen Vectors using MATLAB
CO-2: Develop the code find solution of algebraic and transcendental equations and linear system of equations using MATLAB
CO-3: Write the code to solve problems of first-order ODEs higher order linear differential equations with constant coefficients
CO-4: Fit the curve and visualize all solutions graphically through programmes

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	3	-	-	-	2	-	-	3	2	2
CO-2	3	3	2	2	3	-	-	-	2	-	-	3	2	2
CO-3	3	3	2	2	3	-	-	-	2	-	-	3	2	2
CO-4	3	3	2	2	3	-	-	-	2	-	-	3	2	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1-3:
Eigen Values and Eigen Vectors:
Programs:
- Finding real and complex Eigen values
- Finding Eigen vectors
- SVD and PCA

WEEK 4-5:
Linear System of Equations: Jacobi's iteration method and Gauss-Seidel iteration method
Programs:
- Solution of given system of linear equations using Jacobi's method
- Solution of given system of linear equations using Gauss-Seidel method

WEEK 6-7:
Solution of Algebraic and Transcendental Equations: Bisection method, Newton Raphson Method

Programs:
- Root of a given equation using Bisection method.
- Root of a given equation Newton Raphson Method.

WEEK 8-11:
First and Higher Order ODEs with Constant Coefficients
Exact and non-exact equations, Applications: exponential growth/decay, Newton's law of cooling.
Programs:
- Solving exact and non-exact equations
- Solving exponential growth/decay and Newton's law of cooling problems
- Solving homogeneous ODEs
- Solving non-homogeneous ODEs

WEEK 12-13:
Curve Fitting
Linear and Non-linear curve fitting
- Fitting a straight line
- Fitting non-linear curves

WEEK 14:
Lab Internal Exam

TEXT BOOKS:

1. The Fundamentals of MATLAB: First Programs, Kenneth A. Lambert, Cengage Learnings, 2011
2. Think MATLAB, Allen B. Downey, 1st Edition, Orielly
3. Introduction to MATLAB Programming, William Mitchell, Povel Solin, Martin Novak et al., NCLab Public Computing, 2012
4. Introduction to MATLAB Programming, Jacob Fredslund, 2007

REFERENCES:

1. An Introduction to MATLAB, John C. Lusth, The University of Alabama, 2011
2. Introduction to MATLAB, Dave Kuhlman, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC2CY201) COMPUTER NETWORKS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To understand the fundamental concepts and working principles of computer networks and communication protocols
- To implement data link layer protocols, framing techniques, and error control mechanisms
- To analyze routing algorithms, congestion control techniques, and flow control mechanisms in networks
- To gain hands-on experience with network simulation and analysis tools such as Wireshark and Nmap
- To evaluate network performance metrics such as throughput, delay, packet loss, and congestion

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Implement data link layer framing techniques and error detection mechanisms

CO-2: Analyze and implement error control and flow control protocols in computer networks

CO-3: Implement and analyze routing and congestion control algorithms used in packet-switched networks

CO-4: Apply basic network security, packet analysis, and traffic monitoring techniques using networking tools

CO-5: Design, configure, and analyze computer networks using network simulation and configuration tools

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	2	3	-	-	-	-	-	1	3	1	-
CO-2	3	3	2	2	3	-	-	-	-	-	1	3	2	-
CO-3	3	3	2	3	3	1	2	-	-	-	1	3	2	1
CO-4	2	3	2	3	3	1	1	1	2	-	2	2	3	2
CO-5	3	3	3	3	3	1	1	2	2	1	2	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1: Implement the data link layer framing methods such as character framing, character stuffing, and bit stuffing.

WEEK 2: Implement CRC code computation using CRC-12, CRC-16, and CRC-CCITT polynomials on a given dataset.

WEEK 3: Implement the Sliding Window Protocol and Go-Back-N ARQ mechanism.

WEEK 4: Implement Dijkstra's shortest path algorithm for routing in a computer network.

WEEK 5: Generate and analyze a broadcast tree for a given subnet.

WEEK 6: Implement the Distance Vector Routing algorithm.

WEEK 7: Implement congestion control using the Leaky Bucket algorithm.

WEEK 8: Implement data encryption and decryption techniques for secure communication.

WEEK 9: Demonstrate frame sorting and sequencing techniques used in network buffers.
Week 10: Perform packet capture and analysis using Wireshark, including statistics and filter operations.

WEEK 11: Perform network scanning and operating system detection using Nmap.

WEEK 12 & 13: Configure and simulate network topologies using Cisco Packet Tracer.

TEXT BOOK:

1. Computer Networks, Andrew S. Tanenbaum and David J. Wetherall, 5th Edition, Pearson Education / PHI

REFERENCES:

1. An Engineering Approach to Computer Networks, S. Keshav, 2nd Edition, Pearson Education
2. Data Communications and Networking, Behrouz A. Forouzan, 3rd Edition, TMH

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC2CS201) OBJECT ORIENTED PROGRAMMING THROUGH JAVA LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To understand the fundamentals of Java programming, recursion, and OOP principles to develop modular programs
- To apply multithreading, exception handling, and file handling techniques to create robust and reliable applications
- To analyze and utilize data structures, collections, and generics for efficient data management in Java programs
- To design interactive GUI applications using Swing components, event handling, and basic JDBC connectivity

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Demonstrate programming skills in Java by writing recursive methods, using classes, objects, and packages

CO-2: Implement inheritance, abstract classes, and interfaces to create reusable and extensible programs

CO-3: Develop applications using exception handling, custom exceptions, and multithreading to achieve concurrent execution

CO-4: Construct GUI-based applications with Swing, including calculators, traffic light simulations, menus, and event-driven programs

CO-5: Employ collections, generics, wrapper classes, and JDBC to efficiently manage and manipulate data in Java applications

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	2	-	-	-	-	-	1	3	2	2
CO-2	3	3	3	-	2	-	-	-	-	-	1	3	3	2
CO-3	3	3	2	2	2	-	1	-	-	-	2	2	3	3
CO-4	2	2	3	1	3	-	2	2	-	-	1	2	3	3
CO-5	3	3	2	1	3	-	-	-	-	-	2	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1:

Use Eclipse or Net bean platform and acquaint yourself with the various menus. Create a test project, add a test class, and run it. See how you can use auto suggestions, auto fill. Try code formatter and code refactoring like renaming variables,

methods, and classes. Try debug step by step with a small program of about 10 to 15 lines which contains at least one if else condition and a for loop.

WEEK 2:

Create user-defined packages with multiple classes, Access packages using CLASSPATH, Basic OOP program using classes, objects, constructors

WEEK 3:

a. Write a Java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle, and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.
b. Develop a java application to implement currency converter (Dollar to INR. EURO to INR, Yen) using Interfaces.

WEEK 4:

a. Write a Java program that creates a user interface to perform integer divisions. The user enters two numbers Num1 and Num2. The division of Num1 and Num 2 is displayed as the Result. If Num1 or Num2 were not an integer, the program would throw a Number Format Exception. If Num2 were Zero, the program would throw an Arithmetic Exception.
b. Write a Java program to throw an exception "Insufficient Funds" while withdrawing

WEEK 5 :

a. Write a Java program that implements a multi-thread application that has three threads. First thread generates a random integer every 1 second and if the value is even, the second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number.
b. Write a Java program that correctly implements the producer- consumer problem using the concept of inter thread communication.

WEEK 6:

a. Suppose that a table named Table. txt is stored in a text file. The first line in the file is the header, and the remaining lines correspond to rows in the table. The elements are separated by commas. Write a java program to display the table using Labels in Grid Layout.
b. Write a Java program to list all the files in a directory including the files present in all its subdirectories.

WEEK 7:

a. Create a class called Employee with properties name(String), dateofbirth (java. util. Date), department(String), designation(String) and Salary(double). Create respective getter and setter methods and constructors (no-argument constructor and parameterized constructor) for the same. Create an object of the Employee class and save this object in a file called "data" using serialization. Later using deserialization read this object and prints the properties of this object.
b. Write a Java program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in a record are separated by a tab (\t). It takes a name or phone number as input and prints the corresponding other value from the hash table (hint: use hash tables).

WEEK 8:

a. Create a generic class called Box that can hold any type of object. Implement the following methods: 1) void set(T obj): sets the object stored in the box 2) T get(): retrieves the object stored in the box 3) boolean isEmpty(): returns true if the box is empty, false otherwise
b. Write a Java program to implement Autoboxing and Unboxing.
c. Write a Java program to demonstrate the working of different collection classes. [Use package structure to store multiple classes].

WEEK 9:

a. Write a Java program for the following: Create a doubly linked list of elements. Delete a given element from the above list. Display the contents of the list after deletion.
b. Write a Java program that handles all mouse events and shows the event name at the center of the window when a mouse event is fired (Use Adapter classes).

WEEK 10:

Write a Java program that simulates a traffic light. The program lets the user select one of three lights: red, yellow, or green with radio buttons. On selecting a button, an appropriate message with "Stop" or "Ready" or "Go" should appear above the buttons in the selected color.initially, there is no message shown.

WEEK 11:

a. Write a Java program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -, *, % operations. Add a text field to display the result. Handle any possible exceptions like divided by zero.
b. Develop a Java program that receives an integer in one text field, and computes its factorial Value and returns it in another text field, when the button named "Compute" is clicked.

WEEK 12:

1. Deve1op a Java application to establish a JDBC connection, create a table student with properties name, register number, mark 1, mark2, mark3.insert the values into the table by using the java and display the information of the students at font end.
2. Write a program to perform CRUD operations on the student table in a database using JDBC

WEEK 13:

CBP Evaluation

WEEK 14:

Internal Exam

TEXT BOOKS:

1. Java for Programmers, P. J. Deitel and H. M. Deitel, 10th Edition, Pearson Education
2. Thinking in Java, Bruce Eckel, Pearson Education

REFERENCES:

1. Java Programming, D. S. Malik and P. S. Nair, Cengage Learning
2. Core Java, Volume 1, 9th Edition, Cay S. Horstmann and G Cornell, Pearson

ONLINE RESOURCES:

1. https://codingbat.com/java
2. https://www.w3schools.com/java/java_oop.asp

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC2IT201) DATABASE MANAGEMENT SYSTEMS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Computer Programming Skills

COURSE OBJECTIVES:

- To provide the fundamental concepts of database creation
- To implement the concepts of data manipulation
- To develop procedures for querying multiple tables
- To understand the concepts of PL / SQL

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the given scenario, design it through ER model and normalize the schema

CO-2: Use SQL to construct queries for database creation and manipulation

CO-3: Apply integrity constraints for creating consistent RDBMS environment

CO-4: Applying PL/SQL for processing database

CO-5: Develop solutions for real time requirements of database applications

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	1	-	-	-	-	-	1	2	2	1
CO-2	3	3	2	1	2	-	-	-	-	-	1	2	2	1
CO-3	2	3	2	1	2	-	1	-	-	-	1	2	2	1
CO-4	2	2	2	2	2	-	-	-	-	-	2	2	2	1
CO-5	3	3	3	2	3	-	-	1	1	-	2	2	2	1

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1:

Implement the following using the DUAL table: a) Character functions, b) Numeric functions c) Date functions and d) Conversion functions.

WEEK 2:

Practice DDL and DML commands on a basic table without integrity constraints.

WEEK 3:

Practice DDL and DML commands on a Relational Database, specifying the Integrity constraints. (Primary Key, Foreign Key, CHECK, NOT NULL)

WEEK 4:
Apply the concepts of Joins, SET operations and SQL functions on any two relational schemas

WEEK 5-7:
Apply the concepts of Joins, SET operations and SQL functions on the following schema:

a) Employee:

Name	Datatype	width	Constraint	Description
Empno	Integer	4	Primary Key	Employee Number
Ename	Varchar	20		Employee Name
Job	Char	12		Designation
Mgr	Integer	4		Manager Number
Hiredate	Date			
Sal	Number	(8, 2)		Salary
Comm	Number	(6, 2)		Commission
Deptno	Integer	2	Foreign Key	Department Number

b) Dept:

Name	Datatype	width	Constraint	Description
Deptno	Integer	2	Primary Key	Department Number
Dname	Varchar	12		Department Name
Loc	Char	10		Location

c) Salgrade:

Name	Datatype	width	Constraint	Description
Grade	Integer	1		Grade
Hisal	Integer	4		Upper
Losal	Integer	5		Lower

WEEK 8 – 9:
End to end implementation of a schema for a specific system along with the illustrations of querying.
A system is described by specifying the functional and non-functional requirements. Based on this description, the major entities are identified and modelled. Further the relationships are modelled to form the initial schema. The schema is further refined by removing redundancies through normalization. Also based on the query requirements, the schema is remodeled to facilitate querying. Finally an illustration of various queries to extract required information from the system is shown using SQL / MYSQL.

The five major workflows to be implemented are:
1. System Specification
2. Design of Initial Schema
3. Schema refinement using functional dependencies and normalization
4. Schema refinement using query requirements
5. Illustration of querying the system.

WEEK 10:
Implementation of SQL Window Functions.

WEEK 11:
Implementation of PL / SQL concepts: functions and procedures.

WEEK 12:
Creating and executing CURSORS.

WEEK 13:
Creation and application of TRIGGERS on a Relational schema.

WEEK 14:
Design and implement a user authentication system that employs dynamic SQL queries for validating login credentials. Demonstrate how this approach is vulnerable to SQL injection attacks by bypassing authentication using malicious input. Subsequently redesign the same system to securely handle the user input and prevent SQL injection attacks.

TEXT BOOKS:

1. Database System Concepts, Silberschatz, Korth, Sudarshan, 7th Edition, McGraw-Hill
2. Introduction to Database Systems, C. J. Date, Pearson Education, 1999
3. Database Systems Using Oracle: A Simplified Guide to SQL and PL/SQL, Shah, PHI, 2016

REFERENCES:

1. Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, Tata McGraw-Hill, 2002
2. Fundamentals of Database Systems, Elmasri Navathe, Pearson Education, 2017
3. Database Systems Design, Implementation, and Management, Peter Rob & Carlos Coronel, 7th Edition, Cengage Learning

ONLINE RESOURCES:
1. https://www.w3schools.com/sql/default.asp
2. https://www.javatpoint.com/dbms-tutorial
3. https://www.scaler.com/topics/course/dbms/
4. https://www.geeksforgeeks.org/dbms/dbms/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25PC2IN211) IOT READINESS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To develop practical skills in measuring, estimating, and managing power consumption in IoT hardware systems
- To understand and analyze efficiency, thermal behaviour, and reliability of power regulation and distribution components
- To gain hands-on experience in fault diagnosis, safe load switching, and protection mechanisms used in IoT devices
- To design, assemble, and validate a reliable and deployable IoT power and control module using industry-standard practices

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Measure and analyze power budgets of IoT nodes and evaluate voltage drops and connector reliability in power paths

CO-2: Compare power regulation techniques (linear vs. switching) and assess their efficiency and thermal performance

CO-3: Implement safe control of loads using MOSFETs, relays, and battery charging/protection circuits

CO-4: Integrate, assemble, and test a complete IoT power and control subsystem on a GP board, ensuring safety and reliability

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	3	1	1	3	2	2	2	3	2	3
CO-2	3	3	2	2	3	1	1	3	2	2	2	3	2	3
CO-3	3	2	2	2	3	3	1	3	2	3	2	3	3	3
CO-4	3	3	3	3	3	3	1	3	3	3	2	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1: Power Budget Estimation for an IoT Node Using DC Power Supply and Multimeter

WEEK 2: Breadboard Power Rail Integrity and Voltage Drop Analysis

WEEK 3: Fault Diagnosis in Electronic Circuits Using Digital Multimeter

WEEK 4: Thermal Analysis of Linear Voltage Regulators Under Load Conditions

WEEK 5: Efficiency Comparison Between Linear Regulator and Buck Converter

WEEK 6: Li-Ion Battery Charging and Protection Behaviour Using TP4056 Module

WEEK 7: High-Current Load Switching Using Logic-Level MOSFET Module

WEEK 8: Study of Relay Module with Opto-Isolation and Flyback Protection

WEEK 9: Connector Reliability and Contact Resistance Measurement in Power Lines

WEEK 10: Solder Joint Quality Analysis and Rework Techniques on GP Board

WEEK 11: Automatic Power Path Switching Between Adapter and Battery Supply

WEEK 12: Design and Assembly of a Permanent IoT Power and Control Module on GP Board

TEXT BOOKS:

1. Electronic Devices and Circuits, Bell David A., 5th Edition, Oxford University Press, 2008
2. Electronic Instruments and Measurements, L. D Jones

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25SD5CS201) DATA VISUALIZATION LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To install and run the Rstudio for data analysis
- To understand principles and techniques for data visualization
- To visualize the data that can improve comprehension, communication, and decision making
- To implement various tools and methods for easy interpretation of data

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the importance of data visualization in analytics

CO-2: Gain knowledge in the principles of data visualization

CO-3: Apply the principles of data visualization on toy data sets using R

CO-4: Analyze data towards decision making using visualization

CO-5: Identify appropriate/suitable visualization for particular requirements imposed by the data type and analytics algorithms

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	-	1	-	-	-	1	-	2	2	1	1
CO-2	2	2	2	1	2	-	-	-	1	-	2	3	1	1
CO-3	2	3	2	2	3	-	-	-	1	-	2	3	2	2
CO-4	2	3	2	2	2	1	-	-	2	1	2	3	2	2
CO-5	2	3	3	1	3	-	-	-	2	-	2	3	2	2

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1:

Basics

Introduction to basic components of R programming, overview of visualization, data types, basics of plotting graphs, different types of graphs in analytics

WEEK 2:

Importance of Visualizations

Principles of communicating data, Principles of encoding data to make visualizations, Importance of color in visualizations

WEEK 3:

Plots Using Basic R

Plots with one categorical, continuous variable

Functions in R for plotting, plots with one categorical variable, plots with one continuous variable, plots with one categorical and one continuous variable

Plots with 2 Continuous Variables

Plots with two continuous variables, controlling various aesthetics of the graph**.**

WEEK 4:

ggplot2 in R

Group manipulation and data reshaping in R, understanding the philosophy of ggplot2, bar plot, piechart, histogram, box plot, scatter plot and regression plots

WEEK 5:

ggplot2 in R

Controlling aesthetics like color, size, legend and facets.

WEEK 6:

Data Visualization

Importing the data, Dimensions and measures, color code for various types of variables

WEEK 7:

Working with Sheets

Understanding the worksheet, row and column shelves, showme card, filter and pages shelf

WEEK 8:

Calculations

Working with different measures, creating new calculated fields, Quick table calculations, parameters and groups

WEEK 9:

Graphs for Analytics

Calculate Proportions and percentages, Comparing current to historical and Actual to Target

WEEK 10:

Normal Distribution Variation

Calculate Mean and Median–Normal Distribution Variation and Uncertainty

WEEK 11:

Reporting and Visualizing Variation

Reporting and Visualizing variation, Control Charts Multiple Quantities –Scatter Plots, Stacked bars, Regressions and trend lines

WEEK 12:

Depicting Changes Over Time

Depicting changes over time, Line Chart, Dual Axis Line Chart, scatter plot

WEEK 13:

Reporting

Introduction to dashboard, use of filters in dashboard, Imbedding pictures, Insert live webpage, story

CASE STUDY:

1. Student Performance Analytics Dashboard
2. Sales Performance Analysis for Retail Store
3. Hospital Patient Data Visualization
4. Weather Trend and Climate Analysis
5. Employee Performance and HR Analytics
6. Website Traffic & User Behavior Analysis

WEEK 14

Internal Exam

TEXT BOOKS:

1. Microsoft Power BI, Marco Russo
2. R for everyone, Jarad P Lander
3. Statistics: An Introduction using R, Michael J. Crawley

REFERENCE:

1. Think Python, Allen Downey, Green Tea Press

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. III Semester

(25MN6HS103) HAPPINESS AND WELLBEING

TEACHING SCHEME		
L	T/P	C
2	0	0

EVALUATION SCHEME		
SE-I	SE-II	TOTAL
50	50	100

COURSE OBJECTIVES:

- To learn sustainable strategies to develop positive attitude and happy heart
- To develop self-awareness and self-discipline to meet the needs of happiness
- To practice good health & mindfulness for wellbeing
- To adapt personality attributes of happiness and success strategies
- To nature happiness development index for better living

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Recognize what is happiness in life and how to sustain it

CO-2: Focus on interpersonal skills for a mindful approach

CO-3: Develop to mindfulness to handle challenging situations

CO-4: Recognize the importance of positive attitude for personal and professional development

CO-5: Interpret the need for nurturing happiness development index through Indicators

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	1	1	3	2	2	3	1	3	-	-	-
CO-2	2	1	2	1	1	3	3	2	3	3	3	-	1	-
CO-3	2	1	2	1	1	3	3	2	3	3	3	-	1	-
CO-4	1	1	2	1	1	2	2	3	3	2	3	-	-	-
CO-5	1	1	3	1	1	3	2	3	3	2	3	-	-	1

UNIT-I:

Introduction to Happiness: Definition & theories of happiness: Hedonism theory, Desire theory, Objective list theory. Identifying potential barriers of happiness: Devaluing happiness, chasing superiority, being needy, being overly control-seeking, distrusting others, distrusting life, and ignoring the source within. Strategies for overcoming the potential barriers

UNIT-II:

Power of Emotions & Relationships: Role of emotional intelligence, self-awareness, and empathy in creating harmonious relationship with ourselves and others. Balancing emotions. Hormones that promote happiness. The importance of social connections for happiness. Role of share & care, gratitude, forgiveness & kindness in building relationships

UNIT-III:

Health and Wellbeing: The link between health & happiness-exercise regularly, eat a healthy diet, get enough sleep for physical fitness. Mental wellbeing-Take notice, keep learning, stay connected with nature, and financial wellbeing. The practice of mindfulness and its benefits for mental and physical health. Moving from restlessness to restfulness- meditation and yoga to increase awareness and reduce stress

UNIT-IV:

Re-Wirement for Wellbeing: Abundance in life, freedom of choice, accepting change, ways of implementation for wellbeing: practicing habits-be proactive, begin with end-in-mind, put-first things-first, think win-win, seek first to understand then to be understood, synergize, sharpen the saw, and effectiveness to greatness

UNIT-V:

Nurturing Happiness Development Index: Exploring the sources of temporary joy and lasting happiness. Acceptance, Appreciation, forgiveness, gracefulness, and creative procrastination. Time management with four D's (delete, delay, delegate, do). Developing happiness index-track changes in happiness levels over time and identify the indicators

TEXT BOOKS:

1. The How of Happiness: A Scientific Approach to Getting the Life You Want, Sonja Lyubomirsky, Penguin Books, 2008
2. Authentic Happiness: Using the New Positive Psychology to Realize Your Potential for Lasting Fulfilment, Martin Seligman, Atria Books, 2004
3. The Book of Joy: Lasting Happiness in a Changing World, Dalai Lama, Desmond Tutu, and Douglas Abrams, Avery, 2016

REFERENCES:

1. 7-Habits of Highly Successful People, Stephen Covey, Simon & Schuster, 2020
2. Mindfulness Book of Happiness: Mindfulness and Meditation, Aimen Eman, Publish Drive Edition, 2018
3. Mindfulness at Work: How to Avoid Stress, Achieve More, and Enjoy Life, Dr. Stephen McKenzie, Exisle Publishing, 2014
4. The 8th Habit: From Effectiveness to Greatness, Stephen R. Covey, Free Press, 2004

ONLINE RESOURCES:

1. Life of Happiness and Fulfillment, Indian School of Business, Coursera https://in. coursera.org/learn/happiness
2. Science of Wellbeing, Yale University, Coursera https://www.coursera.org/learn/the-science-of-well-being

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC1AM201) MACHINE LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To introduce the basic concepts and techniques of Machine Learning
- To understand Feature engineering and various probability-based learning techniques
- To have a thorough understanding of the supervised learning techniques
- To learn the concepts of unsupervised learning techniques
- To understand the need for neural networks, deep learning, and reinforcement learning in solving real-time problems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Demonstrate the foundational concepts and principles of machine learning

CO-2: Apply the feature engineering and probability techniques in machine learning

CO-3: Apply supervised learning algorithms, including regression and classification, to real-world problems

CO-4: Interpret the outcomes of unsupervised learning algorithms and convert them into actionable insights for practical applications

CO-5: Apply neural networks, deep learning, and reinforcement learning techniques to real-world case studies for effective problem-solving

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	2	-	-	-	-	-	-	3	2	1
CO-2	3	3	2	2	3	1	-	1	-	-	1	3	3	2
CO-3	3	3	3	2	3	2	1	1	-	2	-	3	3	2
CO-4	2	3	2	3	3	2	1	1	2	2	1	3	3	2
CO-5	3	3	3	2	3	2	2	1	2	2	1	3	3	3

UNIT-I:

Introduction to Machine Learning: Types of Human learning, machine learning process, Well-posed learning problem, Types of machine learning and comparison, applications of machine learning. Model Preparation, Evaluation and feature engineering: Machine learning activities, Types of data in machine learning, dataset understanding, plotting and exploration, checking data quality, remediation, data pre-processing, selecting a model, predictive and descriptive models, supervised learning model training, cross-validation and boot strapping, lazy vs eager learner, interpreting the model- underfitting, overfitting, bias-variance trade-off. Parameter for evaluating the performance of classification, regression, and clustering models. Improving the performance of a model.

UNIT-II:

Feature Engineering: Feature transformation - feature construction, feature extraction by PCA, SVD, LDA. Feature subset selection – feature relevancy and redundancy measures. Feature selection process and approaches. Review of Probability concepts: joint probability, conditional probability, Bayes' rule, Common discrete and continuous distributions, dealing with multiple random variables, and the central limit theorem. Bayes classifier, Multi-class Classification, Naïve Bayes classifier, Bayesian belief network.

UNIT-III:

Supervised Learning - Introduction to supervised learning, Regression: Introduction of regression, Regression algorithms: Simple linear regression, Multiple linear regression, Polynomial regression model, Logistic regression, Maximum likelihood estimation.
Classification: Classification model and learning steps, Classification algorithms: Naïve Bayes classifier, Distance measures, k-Nearest Neighbor (kNN), Decision tree, Support vector machines, Kernel trick, Random Forest.

UNIT-IV:

Unsupervised Learning Introduction to unsupervised learning, Unsupervised vs supervised learning, Application of unsupervised learning, Clustering and its types, Partitioning method: k-Means and K-Medoids, Hierarchical clustering, Density-based methods – DBSCAN

UNIT-V:

Artificial Neural Network: Biological neuron, Artificial neuron, Activation functions, neural network architecture, perceptron, learning process in ANN, Back propagation. Introduction to deep learning, overview of reinforcement learning, Representation learning, Evolutionary learning. Case-study of ML applications: Image recognition, Email spam filtering, Online fraud detection.

TEXT BOOKS:

1. Machine Learning, Saikat Dutt, S. Chjandramouli, Das, 1st Edition, Pearson
2. Machine Learning, M N Murty, Anathanarayana V S, 1st Edition, University Press
3. Machine Learning, Tom M Mitchell, 1st Edition, McGraw-Hill Education, 2013

REFERENCES:

1. Machine Learning – An Algorithmic Perspective, Stephen Marsland, 2nd Edition Chapman and Hall/CRC
2. Machine Learning and Pattern Recognition Series, 2014

ONLINE RESOURCES:

1. https://www.coursera.org/learn/machine-learning
2. https://nptel. ac.in/courses/106/106/106106202/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC1CS205) DESIGN AND ANALYSIS OF ALGORITHMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Programming, Data Structures, Discrete Mathematics

COURSE OBJECTIVES:

- To reinforce algorithms analysis methods
- To analyse running time of an algorithm
- To understand different algorithm design strategies
- To apply algorithms design techniques to solve computational problems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain fundamental algorithmic concepts such as asymptotic notations, recurrence relations, and performance measures of algorithms

CO-2: Apply algorithm design techniques—divide & conquer, greedy, dynamic programming, backtracking, and branch & bound—to solve computational problems

CO-3: Analyze the time complexity of algorithms and compare different solutions using asymptotic bounds

CO-4: Develop graph algorithms to solve real world problems and analyze time complexity

CO-5: Identify computational complexity classes P, NP, NP-Hard and NP complete problems and formulate solutions using standard approaches

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	1	-	-	-	-	-	1	2	2	-
CO-2	3	3	2	1	2	-	-	-	-	-	1	3	3	2
CO-3	3	3	-	2	1	-	-	-	-	-	2	2	3	-
CO-4	3	3	2	2	2	1	-	-	-	-	2	3	3	3
CO-5	2	3	1	2	1	-	-	-	-	-	3	2	3	2

UNIT-I:

Algorithm Analysis: Characteristics of Algorithm. Analysis of Algorithm, Best, Average and Worst-Case behavior, Asymptotic Notations– Order of Growth, Big oh, Omega, Theta notations and Little oh, Little omega notations, Analysis of Recursive Algorithms through Recurrence Relations, Substitution Method, Recursion Tree Method and Masters' Theorem.

UNIT-II:

Divide and Conquer: General method, Applications- Binary search, Quick sort, Merge sort, Strassen 's matrix multiplication.

Greedy Method: General method, Applications- Job sequencing with deadlines, Fractional knapsack problem, Minimum cost spanning trees (Prims and Kruskal), Single source shortest path problem (Dijkstra), Huffman Codes.

UNIT-III:

Dynamic Programming: General method, Principle of optimality, Applications- Matrix chain multiplication, Optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem (Floyd Warshall), Travelling sales person problem, Reliability design. Longest Common Subsequence problem (LCS).

Pattern Matching Algorithms: Brute force, KMP.

UNIT-IV:

Backtracking Method: Applications- N-Queen problem, Sum of subsets problem, Graph coloring, Hamiltonian cycles.

Branch and Bound Method: Applications - Travelling sales person problem, 0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution

UNIT-V:

Graph Algorithms: Depth First Search (DFS) and Breadth First Search (BFS), Topological sorting of DAGs, Connected, Bi Connected and strongly connected components, finding articulation points, Network Flow Algorithm (Ford Fulkerson Algorithm).

Tractable and Intractable Problems: P, NP, NP-complete and NP-hard. Cook's theorem, Reductions: 3-SAT, CLIQUE.

TEXT BOOKS:

1. Fundamental of Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekharam, 2nd Edition, Galgotia Publications
2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Lieserson, Ronald L. Rivest and Clifford Stein, 4th Edition, MIT Press/McGraw-Hill

REFERENCES:

1. Algorithm Design, Jon Kleinberg and Eva Tardos, 1st Edition, Pearson
2. Algorithm Design: Foundations, Analysis, and Internet Examples, Michael T. Goodrich and Roberto Tamassia, 2nd Edition, Wiley
3. Introduction to the Design and Analysis of Algorithms, Anany Levitin, 3rd Edition, Pearson Publications
4. The Design and Analysis of Computer Algorithms, A. Aho, J. Hopcroft and J. Ullman

ONLINE RESOURCES:

1. http://nptel.ac.in/courses/106101060/
2. https://www.cmi.ac.in/~madhavan/teaching.html
3. https://www.ics.uci.edu/~eppstein/161/960312.html
4. https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC1AM101) MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To apply logical reasoning to variety of problems
- To apply Predicate logic reasoning to variety of problems
  To learn the applications on sets, relations and functions
- To implement the concept of counting and formulate and solve recurrence relations
- To analyse the properties of graphs and trees

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Demonstrate problems using Mathematical Logic
CO-2: Predicate logic and set theory
CO-3: Analyze sets, relations, functions, and discrete structures
CO-4: Apply and calculate permutations and combinations
CO-5: Understand the use of graphs and trees as models

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	-	1	-	-	-	-	3	2
CO-2	3	3	1	2	1	2	-	-	-	1	-	-	3	2
CO-3	3	3	1	2	1	2	-	-	2	-	-	-	3	2
CO-4	3	3	1	2	1	2	-	2	-	1	-	-	3	2
CO-5	2	3	2	2	2	2	-	-	-	-	-	-	2	3

UNIT-I:
Mathematical Logic: Statements and notations, Connectives, Well-formed formulas, Truth Tables, tautology, equivalence implication, Normal forms, Rules of inference, Consistency, proof of contradiction.

UNIT-II:
Predicate Logic: The Predicate Calculus, Inference Theory of the Predicate Calculus**.**
Set Theory: Notations, inclusion and equality sets, operations on sets, Venn diagrams, basics of Discrete Structures.

UNIT-III:
Relations: Properties of binary Relations, equivalence, transitive closure, compatibility and partial relations, Hasse diagram.
Functions: Types of Functions, Inverse Function, Composition of functions, recursive Functions.

UNIT-IV:

Counting: Basics of counting, Permutations and combinations - Generalized Permutations and combinations – the Principles of Inclusion– Exclusion, Pigeonhole Principle and its Application.

Recurrence Relations: Generating functions, Generating Functions of Permutations and Combinations.

UNIT-V:

Graph Theory: Basic Concepts, Isomorphism and Subgraphs, Trees and their Properties, Spanning Trees, Directed Trees, Binary Trees, Planar Graphs, Euler's Formula, Multi-graphs and Euler Circuits, Hamiltonian Graphs, Chromatic Numbers, The Four-Color Problem.

TEXT BOOKS:

1. Discrete Mathematical Structures with Applications to Computer Science, J. P. Trembly and R. Manohar, Tata McGraw-Hill, 2019
2. Discrete Mathematics for Computer Scientists & Mathematicians, J. L. Mott, A. Kandel, T. P. Baker, 2nd Edition, PHI, 1986
3. Graph Theory: With Application to Engineering and Computer Science, Narsingh Deo, Prentice Hall of India, 2003

REFERENCES:

1. Elements of Discrete Mathematics, A Computer Oriented Approach, C. L. Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw-Hill, 2017
2. Discrete Mathematics and its Applications, Kenneth H. Rosen, Tata McGraw-Hill, 2003
3. Discrete Mathematics, Norman L. Biggs, 2nd Edition, Oxford University Press, 1989

ONLINE RESOURCE:

1. NPTEL Course on Discrete Mathematical Structures, Prof. Kamala Krithivasan, IIT Madras

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC1IT204) OPERATING SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand operating system overview, functions, services, process management
- To analyse inter process communication and synchronization
- To understand memory management and file system interface
- To understand secondary storage management
- To understand protection and security

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand operating system overview, functions, services and processes

CO-2: Understand inter process communication and synchronization

CO-3: Understand memory management and file system interface

CO-4: Understand secondary storage management

CO-5: Understand protection and security

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	1	2	1	1	1	1	1	-	1	3	3	1	3
CO-2	3	1	2	1	1	1	1	1	-	1	3	3	1	3
CO-3	3	1	2	1	1	1	1	1	-	1	3	3	1	3
CO-4	3	1	2	1	1	1	1	1	-	1	3	3	1	3
CO-5	3	1	2	1	1	1	1	1	-	1	3	3	1	3

UNIT-I:

Operating System Overview: Overview of Computer Operating Systems, Operating System Functions, Operating System Services and Systems Calls, System Programs.

Process Management: Process Concepts, process states, Scheduling-Criteria, CPU Scheduling, Algorithms Evaluation, Threads, multithreading, Thread Scheduling.

UNIT-II:

Inter Process Communication: Message passing, shared memory.

Process Synchronization: Critical-Section Problem, Peterson 's Solution, Hardware support for Synchronization, Mutex Locks, Semaphores, Monitors, Classic Problems of Synchronization- Producer – consumer problem, Dining philosophers' problem, Readers – Writers Problem.

UNIT-III:

Deadlock: Deadlock System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

UNIT-IV:

Memory Management: Memory Management Swapping, Contiguous Memory Allocation, Paging, Page-Table Structure, Segmentation, Virtual Memory, Demand Paging, Page-Replacement Algorithms, Frames Allocation, Thrashing.

File System Interface: File Concepts, Access Methods and Directory Structure, File System Mounting, File Sharing and Protection, Allocation Methods, Free-Space Management, Efficiency and Performance.

UNIT-V:

Mass Storage: Overview Mass-Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, Swap-Space Management, RAID Structure

Protection: Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of Access Matrix,

Security: Security Problem, Program Threats, System and Network Threats

TEXT BOOKS:

1. Operating System Concepts, Abraham Silberchatz, Peter B. Galvin, Greg Gagne, 10th Edition, John Wiley
2. Operating Systems- A Concept based Approach, D. M. Dhamdhere, 2nd Edition, Tata McGraw-Hill

REFERENCES:

1. Operating Systems - Internal and Design Principles, Stallings, 5th Edition, Pearson Education/PHI, 2005
2. UNIX: Concepts and Applications, Sumitabha Das, 4th Edition, Tata McGraw-Hill
3. Operating System: A Design-Oriented Approach, Charles Patrick Crowley
4. Operating Systems: A Modern Perspective, Gary J., Nutt
5. Design of the Unix Operating Systems, Maurice J., Bach

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC1IT208) COMPUTER ORGANIZATION AND ARCHITECTURE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:
- To understand the functional blocks of computer and instruction set architecture
- To design basic computer organization and addressing modes of Central Processing Unit
- To understand micro programming and to perform arithmetic micro operations on integers and floating-point numbers
- To discuss the different ways of communicating with I/O devices
- To design techniques to enhance the performance using pipelining, parallelism

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the basic components and the design of CPU, instruction set
CO-2: Design basic computer organization and to understand instruction formats and addressing modes
CO-3: Understand the micro programming and Implement arithmetic operations on integer and floating type values
CO-4: Analyze and understand the input organization and memory organization
CO-5: Explore pipeline processing, vector processing and array processing

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	-	1	-	-	-	-	-	1	3	1	-
CO-2	3	3	2	1	1	-	-	-	-	-	1	3	1	-
CO-3	3	3	2	2	2	-	-	-	-	-	2	3	2	-
CO-4	3	3	1	2	2	-	-	-	-	-	2	3	2	-
CO-5	3	2	2	1	2	-	-	-	-	-	2	3	3	1

UNIT-I:
Functional Blocks of a Computer: Introduction, Block Diagram, Definition of Computer Organization, Computer Design and Computer Architecture
Data Representation: Data types, Complements, Fixed Point Representation, Floating Point Representation.
Instruction Set Architecture: Register Transfer language, Register Transfer, Arithmetic Micro operations, logic micro-operations, shift micro-operations, Arithmetic logic shift unit.

UNIT-II:
Basic Computer Organization and Design: Instruction codes, Computer Registers Computer instructions, Timing and Control, Instruction cycle, Memory Reference Instructions, Register Reference and Input – Output.

Central Processing Unit: General Register Organization, Instruction Formats, Addressing modes, Data Transfer and Manipulation.

UNIT-III:

Microprogrammed Control: Control memory, Address sequencing, micro program example, design of control unit.

Computer Arithmetic: Addition and subtraction, multiplication Algorithms, Division Algorithms, Restoring and Non-restoring, floating – point Arithmetic operations.

UNIT-IV:

Input-Output Organization: Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Program Controlled, Priority, Interrupt, Direct memory Access. Privileged and Non-Privileged Instructions.

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary memory, Associate memory, Cache Memory, Mapping Techniques, Replacement Algorithms, Write Policies, Memory interleaving.

UNIT-V:

Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processor.

TEXT BOOKS:

1. Computer System Architecture, M. Morris Mano, 3rd Edition
2. Computer Organization and Embedded Systems, Carl Hamacher, 6th Edition, McGraw-Hill

REFERENCES:

1. Computer Organization and Design: The Hardware/Software Interfaces, David A. Patterson and John L. Hennessy, 5th Edition, Elsevier
2. Computer Architecture and Organization, John P. Hayes, 3rd Edition, WCB/McGraw-Hill
3. Computer Organization and Architecture: Designing for performance, William Stallings, 10th Edition, Pearson Education
4. Computer System Design and Architecture, Vincent P. Heuring and Harry F. Jordan, 2nd Edition, Pearson Education

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PW4CY201) DESIGN THINKING

TEACHING SCHEME		
L	T/P	C
1	2	2

EVALUATION SCHEME		
CIE	SEE	TOTAL
40	60	100

COURSE OBJECTIVES:

- To instill a sense of significance towards applying creativity to product and service design
- To teach a systematic approach to identifying and defining a problem before brainstorming for a solution
- To inculcate core design principles and applied creativity to develop innovative strategies that better connect engineers and technologies with their end users
- To build a mindset leading to flow of creative ideas, validating those ideas and prioritizing the best ones among them
- To motivate students to apply design thinking while implementing projects focusing on local, regional or global societal problems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Demonstrate the understanding of design principles from a technology perspective

CO-2: Validate problem statements through user empathisation with societal, cultural, global and Environmental consciousness

CO-3: Use specific and relevant ideation and brainstorming techniques to find innovative solutions

CO-4: Prototype a solution to address user challenges

CO-5: Investigate the cultural, emotional, environmental, technological and business factors relevant to developing new product or service design concept

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	-	2	-	-	2	-	-	3	3	1	2	-	-
CO-2	-	-	3	2	1	3	-	-	3	2	1	1	-	-
CO-3	-	-	2	-	-	2	-	-	2	2	3	1	-	-
CO-4	-	-	3	-	2	-	-	-	2	3	1	2	3	3
CO-5	-	-	-	3	-	3	1	-	1	-	1	1	1	1

UNIT-I:

Design Overview: Need of Design in Engineering, Various perspectives of design, Case study of Design thinking at Apple, IBM, IDEO and GE Healthcare with lessons learned on Good and Bad Design, Human-Centred Design (HCD) - Desirability, Feasibility, viability. Problem Identification - 17 SDGs as the core thematic problem areas.

Design Process: Introduction to various Design models, The Design Double Diamond model: Discover-Define-Develop-Deliver and Stanford's d. school design thinking model: Empathy-Define-Ideate-Prototype-Test.

UNIT-II:

User Empathy: Identifying Stakeholders and Defining User Personas, Data collection-User Interviews and conducting surveys, Empathy Tools: What/Why/How, Five Whys, Empathy Maps, AEIOU, Customer Journey Maps (CJM).

Need Analysis: Types of Users, making sense of observations and insights, Types of Needs-Primary, secondary, latent needs, Value Proposition to the Users, Identifying Addressable Needs and Touch points, Structuring Need Statements.

UNIT-III:

Ideation Process: Introduction to creativity and closed-world solutions, Idea generation techniques: Brainstorming, Mind Maps, SCAMPER, Systematic Inventive Thinking methods (Subtraction, Multiplication, Division, Task Unification, Attribute Dependency), 3-box thinking, and Box-3 ideation.

UNIT-IV:

Building Prototypes & Design for Sustainability: Building Conceptual model of product/service using various prototype methods. Concern for Environment and Sustainability in Design, Case Studies to understand Design for Environment (DFE).

UNIT-V:

Capstone Project (Interdisciplinary): Applying design thinking principles and methods for problem definition, ideation, prototyping, testing, refining and taking the solution to the users, using visual representation tools to indicate problem, User persona, needs, empathisation, ideas and prototype that leads to chosen solution, creating presentation.

TEXT BOOKS:

1. Change by Design, Tim Brown, Harper Business, 2012 (ISBN: 978-0062337382)
2. The Design of Everyday Things, Donald A. Norman, MIT Press, 2013 (ISBN: 978-0262525671)

REFERENCES:

1. The Art of Innovation, Tom Kelly, Jonathan Littman, HarperCollins Business, 2002
2. Design Thinking: Integrating Innovation, Customer Experience, and Brand Value, Thomas Lockwood, Allworth Press, 2009 (ISBN: 978-1581156683)
3. Design Thinking for Start-ups: A Handbook for Readers and Workbook for Practitioners, Jimmy Jain, Notion Press, 2018 (ISBN: 978-1642495034)

ONLINE RESOURCES:

1. https://eict.iitg.ac.in/course_img/76/052014859ef93ebe8c4dd4285fdd5dbe09296b.pdf
2. https://www.ideou.com/collections/design-thinking-courses?srsltid=AfmBOoqcLf7_8mqtRd-KMqVCt5J6z6A6ABSbKQNJcvQKFzdq84E1dnjj

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC2AM201) MACHINE LEARNING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To gain an overview of various supervised machine learning techniques
- To understand and evaluate various unsupervised machine learning techniques
- To implement and demonstrate various libraries used in machine learning
- To understand the process of training, building, and evaluating machine learning models
- To apply machine learning techniques to real-world case studies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Illustrate and implement supervised machine learning algorithms

CO-2: Develop and apply unsupervised machine learning algorithms, such as clustering and dimensionality reduction, to analyse patterns in data

CO-3: Explore the various libraries used in Machine Learning

CO-4: Build, train, and evaluate machine learning models using appropriate performance metrics, cross-validation, and model validation techniques

CO-5: Perform experiments on real-world datasets using machine learning techniques

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	3	1	-	1	1	1	-	3	2	2
CO-2	3	3	2	3	3	1	1	1	1	1	1	3	3	2
CO-3	2	2	1	1	3	-	-	-	1	1	-	3	2	2
CO-4	3	3	3	3	3	1	-	1	1	1	1	3	3	3
CO-5	3	3	2	2	3	2	1	1	2	2	1	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1: Write a python program to compute Central Tendency Measures: Mean, Median, Mode, Measure of Dispersion: Variance, Standard Deviation

WEEK 2: Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy

WEEK 3: Study of Python Libraries for ML application such as Pandas and Matplotlib

WEEK 4: Write a Python program to implement Simple Linear Regression

WEEK 5: Implementation of Multiple Linear Regression for House Price Prediction using sklearn

WEEK 6: Implementation of Decision tree using sklearn and its parameter tuning

WEEK 7: Implementation of KNN using sklearn

WEEK 8: Implementation of Logistic Regression using sklearn

WEEK 9: Implementation of K-Means Clustering

WEEK 10: Performance analysis of Classification Algorithms on a specific dataset (Mini Project)

WEEK 11: Implementation of real-world case study - Lower Back Pain Symptoms Clustering/ Personalized Cancer Diagnosis**.**

WEEK 12: Lab Internal Exam

TEXT BOOK:

1. Machine Learning, Tom M. Mitchell, McGraw-Hill

REFERENCE:

2. Machine Learning: An Algorithmic Perspective, Stephen Marshland, Taylor & Francis

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25PC2AM211) OPERATING SYSTEMS WITH LINUX LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To familiarize UNIX/Linux operating system environment and basic command-line operations for effective system usage
- To develop programming skills using UNIX/Linux system calls and shell scripting for process management, file handling, and directory operations
- To provide hands-on experience in simulating core operating system concepts such as CPU scheduling, memory management, deadlocks, and synchronization
- To understand and implement resource allocation and management algorithms related to memory, files, and disk scheduling
- To enhance problem-solving abilities through implementation and analysis of OS algorithms using C programming and shell scripts

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Use basic UNIX/Linux commands and system calls to perform file, process, and directory operations effectively

CO-2: Develop shell scripts to automate tasks, manage files, and analyze permissions in a UNIX/Linux environment

CO-3: Implement and simulate CPU scheduling algorithms and synchronization mechanisms using appropriate programming constructs

CO-4: Design and analyze solutions for deadlock handling, memory allocation, and page replacement algorithms

CO-5: Simulate and evaluate file allocation and disk scheduling algorithms to understand storage management in operating systems

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	3	1	-	2	-	-	-	2	2	2	2
CO-2	3	-	1	2	2	-	2	-	-	-	1	2	3	1
CO-3	3	2	1	3	1	-	1	-	-	-	-	3	2	1
CO-4	3	2	2	2	1	-	-	-	-	-	-	3	1	2
CO-5	3	2	1	2	1	-	-	-	-	-	-	2	2	2

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1: Working with basic UNIX/ Linux commands.

WEEK 2: Programs using the following system calls of Unix / Linux operating system: fork, exec, getpid, exit, wait, close, stat, opendir, readdir

WEEK 3: Programs using the I/O system calls of UNIX operating system (open, read, write)

WEEK 4: Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.

WEEK 5: Simulations of Unix / Linux commands like ls, grep, etc.

WEEK 6: Simulate the following CPU scheduling algorithms a) FCFS b) SJF c) Priority d) Round Robin based algorithm.

WEEK 7: Implementation of synchronization problems using Semaphore.

WEEK 8: Simulate algorithms for deadlock avoidance and deadlock detection.

WEEK 9: Simulate the following Memory Allocation algorithms a) Best fit b) worst fit c) first fit

WEEK 10: Simulate the following page replacement algorithms a) FIFO b) optimal c) LRU

WEEK 11:
Simulate the following file allocation strategies
a) Sequential b) Indexed c)Linked

WEEK 12:
Simulate the following disk scheduling algorithm
a) FCFS b) SCAN c) CSCAN d)LOOK e)CLOOK

TEXT BOOKS:

1. Unix System Programming using C++, T. Chan, PHI
2. Operating System Principles, Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley

REFERENCES:

1. Beginning Linux Programming, 4th Edition, N. Matthew, R. Stones, Wrox, Wiley India Edition, 2008
2. Linux System Programming, Robert Love, O'Reilly, SPD
3. Modern Operating Systems, Andrew S. Tanenbaum, 2nd Edition, Pearson/PHI
4. Operating Systems-A concept Based approach, D. M. Dhamdhere, 2nd Edition
5. Mastering Unix Internals and Shell Scripting, Paperback, Mr. Bhupesh Deka, Dr. Srinivas Prasad, Ms. Bindu Agarwalla

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25SD5CS202) WEB APPLICATION DEVELOPMENT

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Object-Oriented Programming through Java, HTML and CSS Basics, Fundamental understanding of web technologies

COURSE OBJECTIVES:

- To build responsive web pages using HTML5, CSS3, Bootstrap, and modern layout techniques (Flexbox, Grid)
- To implement client-side interactivity and state management using ES6+ JavaScript
- To develop REST APIs and backend services using Node. js and Express. js
- To design and manage NoSQL databases using MongoDB and implement database operations from the backend and build single-page applications (SPAs) with React, including routing, state management, and component lifecycle
- To implement authentication, authorization, and security best practices in full-stack applications and deploy and host full-stack web applications on cloud platforms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Build responsive, accessible web pages using HTML5, CSS3, Bootstrap, and modern CSS layout techniques and Implement client-side interactivity, form validation, and state management using ES6+ JavaScript features

CO-2: Design and develop REST APIs and server-side logic using Node. js and Express. js frameworks and Design NoSQL database schemas, implement CRUD operations, and manage data persistence using MongoDB

CO-3: Build single-page applications with component-based architecture, routing, and state management using React

CO-4: Implement user authentication, authorization, and security mechanisms (JWT, hashing, CORS) in web applications

CO-5: Deploy, test, and host full-stack applications on cloud platforms with CI/CD pipelines

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	3	-	-	1	2	-	2	3	2	1
CO-2	3	3	3	2	3	-	-	1	1	-	2	3	3	2
CO-3	3	3	3	2	3	-	-	2	2	-	2	3	3	2
CO-4	2	3	3	2	3	1	2	1	1	-	2	3	2	2
CO-5	2	3	3	2	3	1	2	2	2	2	3	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1: Responsive Multi-Page Website
Build a responsive website (3-4 pages) for your chosen project domain using HTML5, CSS3, Bootstrap, and modern layout techniques. Pages should include: landing page, content pages, contact/form page. Implement CSS Grid, Flexbox, and responsive design principles.

WEEK 2: Client-Side Form Validation & Interactivity
Add JavaScript interactivity to Exercise 1. Implement client-side form validation, dynamic DOM manipulation, event handling, and local storage. Create interactive UI elements (modal dialogs, collapsible sections, animated transitions).

WEEK 3: Learn ES6+ Features & Fetch API
Explore ES6+ features (arrow functions, destructuring, spread operator, async/await). Create a practice application that fetches data from a public API (OpenWeatherMap, JSONPlaceholder, etc.) and displays it dynamically on the webpage. Implement error handling and loading states.

WEEK 4: RESTful API Development
Develop a REST API for your project using Node. js and Express. js. Create endpoints for key entities in your domain (e. g., products, users, posts, students). Implement GET, POST, PUT, DELETE operations. Use Postman to test all endpoints. Document API with Swagger/OpenAPI

WEEK 5: Error Handling, Validation & Middleware
Enhance your API from Exercise 4 with:

- Request validation middleware
- Error handling middleware
- Logging middleware
- Authentication middleware (basic)

Implement comprehensive error responses and HTTP status codes.

WEEK 6: MongoDB Schema Design & CRUD Operations
Design a MongoDB schema for your project entities. Implement CRUD operations using Mongoose. Create seed scripts to populate sample data. Test all operations. Implement data validation and indexing for performance.

WEEK 7: Integrate Backend API with Express & MongoDB
Connect your Express API (Exercise 4-5) with MongoDB database (Exercise 6). Replace mock data with actual database operations. Implement data persistence, retrieval, and updates through the API.

WEEK 8: React Component Architecture & Hooks
Create a React application with component-based architecture. Implement functional components, hooks (useState, useEffect), props, and component composition. Build interactive UI components for your project (forms, cards, lists).

WEEK 9: React Routing & State Management

Implement React Router for multi-page navigation in your React app. Add route parameters and nested routes. Implement state management using Context API or Redux for shared state across components.

WEEK 10: Integrate React Frontend with Backend APIs

Connect your React SPA (single Page Application) to your backend API. Implement data fetching, form submissions, and CRUD operations through API calls. Add loading states and error handling. Implement pagination and filtering where applicable.

WEEK 11: JWT Authentication & Authorization

Implement user authentication using JWT. Create login/signup endpoints in backend. Implement JWT token generation and validation. Add protected routes in React that require authentication. Implement role-based access control.

WEEK 12: Security Best Practices & Input Validation

Implement security measures:

- Password hashing (bcrypt)
- CORS configuration
- Input validation and sanitization
- Prevention of common vulnerabilities (XSS, CSRF, SQL injection)
- Secure environment variable management

WEEK 13: Deployment & DevOps

Deploy your complete full-stack application to a cloud platform (Heroku, Vercel, Firebase, AWS). Implement Docker containerization. Set up CI/CD pipeline (GitHub Actions). Configure environment-specific configurations. Implement monitoring and logging.

WEEK 14: Capstone Project Submission & Presentation

Present your complete full-stack application demonstrating:

- Responsive frontend with intuitive UI/UX
- Functional backend APIs with proper error handling
- Secure authentication and authorization
- Database operations and data persistence
- Deployment and live hosting
- Code quality and documentation

CAPSTONE PROJECT THEMES (Students Choose One)

1. E-commerce Platform: Product catalog, shopping cart, checkout, order management, user authentication
2. Blog/Content Management System: Article creation/editing, commenting, tagging, search functionality
3. Social Network: User profiles, posts/feeds

TEXT BOOKS:

1. Full Stack Web Development with React, Node. js, Express, and MongoDB, Vasan Subramanian, Apress, 2023

2. Learning Full-Stack JavaScript Development: MongoDB, Node and React, Daniel Shiffman, O'Reilly Media, 2020

REFERENCE:

1. The MERN Stack Tutorial: Building Web Applications with MongoDB, Express, React, and Node, Jack Herrington, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. IV Semester

(25VA7HS201) ANCIENT WISDOM

TEACHING SCHEME		
L	T/P	C
1	0	1

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To provide a fundamental understanding of ancient wisdom from various world civilizations and its relevance to STEAM fields about certain aspects of culture, heritage, and history as sources of knowledge and application for the present and future
- To explore the historical context and contributions of ancient Indian and other global civilizations to STEAM disciplines and to foster scientific temperament using the approaches that stood the tests of time
- To develop critical thinking and logical reasoning by understanding ancient texts, philosophies, and practices, and by exploring their applicability for responsible consumption and sustainable development
- To create a karmic, sustainable, humane, and responsible future for the world with scientific temperament and a sense of belongingness and ownership to our heritage as humans

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe important contributions of ancient Indian and other world civilisations to STEM disciplines, fostering a scientific temperament

CO-2: Conceptually analyse ancient texts and philosophies to identify aspects of sociocultural relevance to the present day with a focus on sustainable development

CO-3: Apply principles, concepts, approaches, and ideas of ancient wisdom to improve human life/ society through STEM professional activities and management

CO-4: Know the importance of learning from the past and developing systems thinking to look at the big picture, leading to interdependence that promotes interdisciplinary approaches to holistic development and ethical future

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	-	1	2	1	-	1	-	2	2	1	1
CO-2	2	3	2	2	-	3	2	-	1	-	2	1	2	1
CO-3	2	3	3	2	1	3	2	1	1	1	2	2	1	2
CO-4	1	2	2	1	-	2	3	2	2	1	3	3	1	1

ACTIVITIES: Dialogue-based lectures, Videos, Debates and discussions, collaborative learning activities and group work including presentations and posters

MODULE 1:

Introduction and Motivation

Introduction: How ancient is ancient? Is all ancient wisdom fundamentally religious or spiritual? Does ancient wisdom still have anything important to remember and consider relevant today? Why should we look back while living in the present and building for the future? How can we separate religion, spirituality and philosophy?

The Indian Story: Was India really a great country in the past? What has changed in India since, and what has not? How is culture and heritage important or relevant for the future that is not like the past at all? Was knowledge really codified in literature? Is Hinduism a way of life or a religion and should we care? Is the caste system or varna system divisive?

History of Sciences and History of Arts: What has humanity achieved so far in Sciences and in Arts? Understanding the geography of thought and pursuit of knowledge; Leading from the emerging future through the history of humanity

Definitions and classifications: Eastern, Arabic, and Western philosophical approaches; Epistemology, Phenomenology, Hermeneutics, Metaphysics, tarka, nyäya, mïmämsa, ikigai, oubaitori, eudaimonia, gestell

MODULE 2:

Deep Dive into the India Story

Ancient India: Indian Ethos, Concepts of dharma, karma, chaturvidha purushaartha (dharma, artha, käma, moksha), triguna (sattva, rajas, tamas) – relevance and interpretation for contemporary living; Major Indian contributions: To Education, Language, Governance, and Academic Disciplines with particular focus on STEAM (STEM and Arts)

Documented and Undocumented Knowledge: Vedas, Vedangas, Upanishats, Epics and Mythology: Ramayana, Mahabharata and Bhagavad-Gita, Puranas including Bhagavata; Examples of codified knowledge from various written verses: did they actually mean what we now interpret them to be?

Medieval India: Advancements and developments in education, governance, technology, and society

Modern India: Constitutional values of India influenced by Ancient and Medieval Indian history and culture

Summary of Values in Indian Ethos: Family system, the karma principle and the idea of nishkaama karma, practising and swadharma, protecting dharma, Endurance and Patience (sahanam), Motivation and Determination (sankalpam), Attention and Devotion (Sraddha), Industriousness and Exertion (Srama), Understanding the Self and Ego, Peace (Saanti), Proactive non-violence (ahimsa), Empathy, Altruism, Focus on the big picture, Inquisitiveness and Pursuit of Knowledge, Harmony, Oneness of Humanity (vasudhaika kutumbam), Learning from the Nature and concern towards other lifeforms

Looking beyond the "proud claims": Myths and facts about Indian ancient wisdom: Pursuing Truth, in the Indian way; Can India be the "Viswa Guru"? If yes, how? If not, can/shall we still work towards it?

MODULE 3:

Ancient Civilisations and Wisdom from other parts of the World

Overview: Indus Valley and other ancient cultures of the Indian subcontinent – their influence on Southeast Asia; Civilisations outside of Indian subcontinent:

Mesopotamian, Egyptian, Far East, European, Middle East and African, North, Central and South American, Oceanic cultures and their indigenous peoples

Advancements of the Ancient World: Advances in governance, arts, medicine and health, architecture, metallurgy, weaponsmithy, mechanisation, cosmology, other science and technology, mathematics, astronomy, language, trade and travel, philosophy, other cultural aspects

Sources of Wisdom, Religion, and Spirituality: Overview of major Abrahamic religions and their holy books: The Holy Qur'an, Hadith, and Sharia for Islam; The Old and New Testaments for Christians; Overview of other world religions and their tenets; Overview of Ancient Greek and Ancient Chinese philosophies; Philosophies of Abrahamic religions and non-Abrahamic religions of Indian origin

MODULE 4: Applicability of Ancient Wisdom

Understanding limitations of ancient wisdom; Examining ancient wisdom to find logical relevance and applicability to the present world; Applying tarka Sästra principles—dialectics, debates, analyses, discussions—and understanding the spirit of scholars of tarka Sästra, such as Adi Sankara, Aristotle, Ramanuja, Plato; Identifying core principles, approaches, and ideologies in formulation, design, strategy, and execution in various STEAM domains

MODULE 5: Lessons to Learn from Ancient Wisdom

Understanding Bhagavad Gita as a text of introspection and motivation, for personality development, team management, and organisational excellence; Exploring ancient books for strategy and management: The Art of War (Sun Tzu), Artha Sastra (Kautilya), parts of ämukta mälyada (Krishna Deva Raya) and Mahabharata (Vyasa/ Tikkana)

The Way Forward: Developing a systems thinking approach; Developing a humanitarian and scientific temperament based on cultural and constitutional values; Leveraging education to resolve societal problems and creating impact with focus on responsible consumption and sustainable development; reinstalling "human" at the centre of development and using technology for enablement instead of enslaving; Understanding the value of interdependence and Presencing

TEXT BOOKS:

1. The Wonder That Was India: Volume I, A. L. Basham, Picador India, 2019 (ISBN: 978-9389109344)
2. Indian Science and Technology in the Eighteenth Century, Dharampal, Rashtrotthana Sahitya, 2021 (ISBN: 978-8175310933)
3. A History of Science in World Cultures: Voices of Knowledge, Scott L. Montgomery and Alok Kumar, Routledge, 2015 (ISBN: 978-0415639842)
4. Sapiens: A Brief History of Humankind, Harper, Yuval Noah Harari, 2015 (ISBN: 978-0099590088)

REFERENCES:

1. The Geography of Thought: How Asians and Westerners Think Differently and Why, Richard E. Nisbett, Nicholas Brealey Publishing, 2004 (ISBN: 978-1529309416)
2. The Question Concerning Technology, and Other Essays, Martin Heidegger, Harper Perennial Modern Classics, 2013 (ISBN: 978-0062290700)

3. Arise Awake & Don't Stop!, Swami Vivekananda, Namaskar Books, 2022 (ISBN: 978-9355717221)
4. The Hindu Way: An Introduction to Hinduism, Shashi Tharoor, Aleph Book Company, 2019 (ISBN: 978-9388292856)
5. Eastern Religions and Western Thought, Sarvepalli Radhakrishnan, Oxford University Press, 1997 (ISBN: 978-0195624564)